

The Evolution of Web Protocols

In this second article in the series on 'Evolution of Web protocols', we follow the development of the HTTP and SSL protocols. We begin with a timeline that depicts the evolution of these protocols.

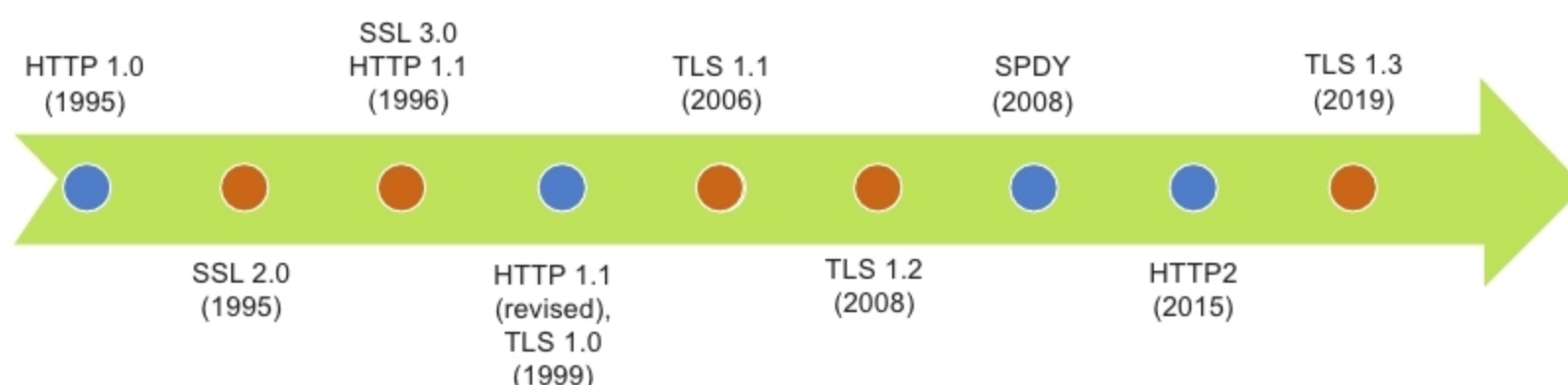


Figure 1: TLS (Transport Layer Security) timeline

In the first article in the series carried in the February issue of *OSFY*, we discussed the evolution of the Web protocol stack since its inception in the 1990s to the present day. The early stack comprised the DNS and HTTP/1.1 application protocols, transported respectively on UDP and TCP, with both the protocols using the services of IP. Both DNS and HTTP were unsecured protocols that could be 'read' by anyone with access to either the network path or a copy of the packet flow. As the Internet grew and was found suitable for commercial applications, this lack of security was not acceptable. Hence, a new session layer protocol called SSL, which encrypted all application traffic, was proposed and eagerly adopted, especially for HTTP, giving rise to the now ubiquitous HTTPS. In the first article in this series, we followed this evolution and looked at the key characteristics of the DNS and HTTP protocols.

Now, let us follow the evolution of the HTTP and SSL protocols to their next versions, starting with a timeline that depicts the evolution of these protocols (Figure 1).

As can be seen in the timeline, HTTP/1.1 remained unchanged for a long period of 16 years (from 1999 to 2015). In the interim, its successor started as SPDY within Google, getting deployed within the Google browser and server stacks. It was then brought to IETF, where it was adopted with relatively minor changes as HTTP2. On the other hand, SSL underwent five revisions between 1995 and 2008, when TLS 1.2 was adopted. TLS 1.2 ruled the Web for a decade before TLS 1.3 was proposed.

What drives Web protocols to evolve

In today's world, the Internet is not just a novelty or a luxury – it is a necessity for a significant fraction of the population. For that reason, it needs to be highly available, secure and fast. While higher availability can be addressed by replacing redundant hardware and cabling, security and speed must be built all through the stack, including into the hardware and software. As we discussed in the last article, the network stack is the foundation of the Internet and it implements the protocols that govern network communication.

While the early network protocols were focused on 'getting it right', the current motivation is to go beyond the functional requirements, and forms the basis for overhauling the protocols for security and performance.

Characteristics of contemporary Web portals

With the emergence of Web 2.0, the nature and quantum of the contents on Web portals changed. We have all experienced that — while the early Web was quite 'bare' and text-intensive, today's portals are much more captivating with images, videos and the unavoidable ads. With more expressive sites, each portal is getting heavier in terms of the number of objects that have to be delivered to the browser. These objects include CSS, images and JavaScript, in addition to the regular HTML files.

For example, when I tried www.flipkart.com from my browser, captured the resultant traffic as a HAR file and analysed it, I observed the following. The browser made a total of 87 HTTP transactions, spread across 10 hosts. The following table summarises my observations (note